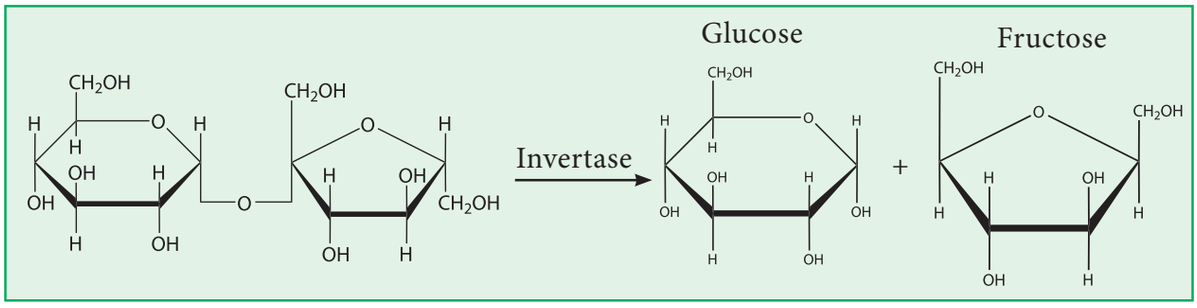
i) Controlled reduction of sucrose gives a mixture of sorbitol and mannitol.

j) It reacts with lime water $Ca(OH)_2$ to give calcium sucrate.

Hydrolysis of sucrose:

Sucrose is hydrolysed by dilute acids or enzymes like sucrase or invertase into an equimolar mixture of glucose and fructose.

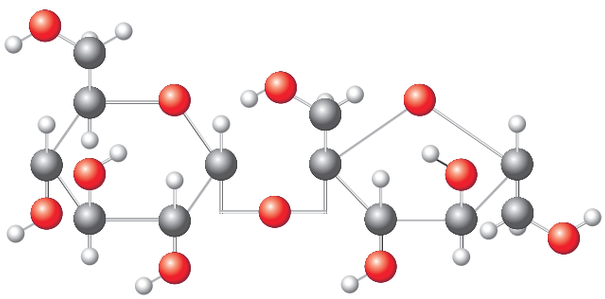


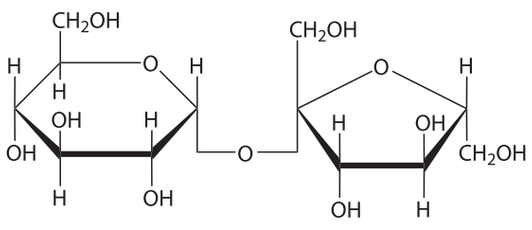
Sucrose is dextrorotatory. But the hydrolysed product is laevorotatory. Since the direction of rotation is reversed, this phenomenon is known as inversion of cane sugar. The mixture of sugars formed on hydrolysis is known as invert sugar.

According to Hudson, sucrose is first split into α-D(+) glucopyranose and β-D(+) fructofuranose, both are dextro rotatory. However, the less stable β-D(+) fructofuranose then sets up an equilibirum with its more stable isomer, α-D(–) fructopyranose which is strongly laevorotatory. Thus, the invert sugar gives a specific rotation of –28.2°.

Structure of Sucrose:

Sucrose (saccharose)





α-D-glucopyranosyl-β-D-fructofuranoside

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